Experiment-8

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Course Code: -MLA0305 **Slot-**B

Title

Comparative Analysis of SARSA and Q-Learning for Reinforcement Learning

Aim

To implement and compare the **SARSA** and **Q-Learning** algorithms for solving a Reinforcement Learning problem and analyze their performance based on reward, convergence, success rate, and optimal policy.

Procedure

1. Define a Reinforcement Learning environment with a finite set of states and actions.
2. Define the transition function and rewards for each state-action transition.
3. Initialize separate **Q-tables** for SARSA and Q-Learning.
4. Set the learning rate , discount factor , and exploration rate .
5. Use an **ε-greedy policy** to balance exploration and exploitation.
6. For **SARSA**:
    - Select the current action using the ε-greedy policy.
    - Execute the action and observe the next state and reward.
    - Select the next action using the same policy.
    - Calculate the TD target:

      $$Rt + 1 + \gamma Q(St + 1, At + 1)$$

    - Update the Q-value using:

      $$Q(St, At) \leftarrow Q(St, At) + \alpha[Rt + 1 + \gamma Q(St + 1, At + 1) - Q(St, At)]$$

7. For **Q-Learning**:
    - Select an action using the ε-greedy policy.
    - Execute the action and observe the next state and reward.
    - Select the maximum Q-value of the next state.
    - Calculate the TD target:

      $$Rt + 1 + \gamma amaxQ(St + 1, a)$$

    - Update the Q-value using:

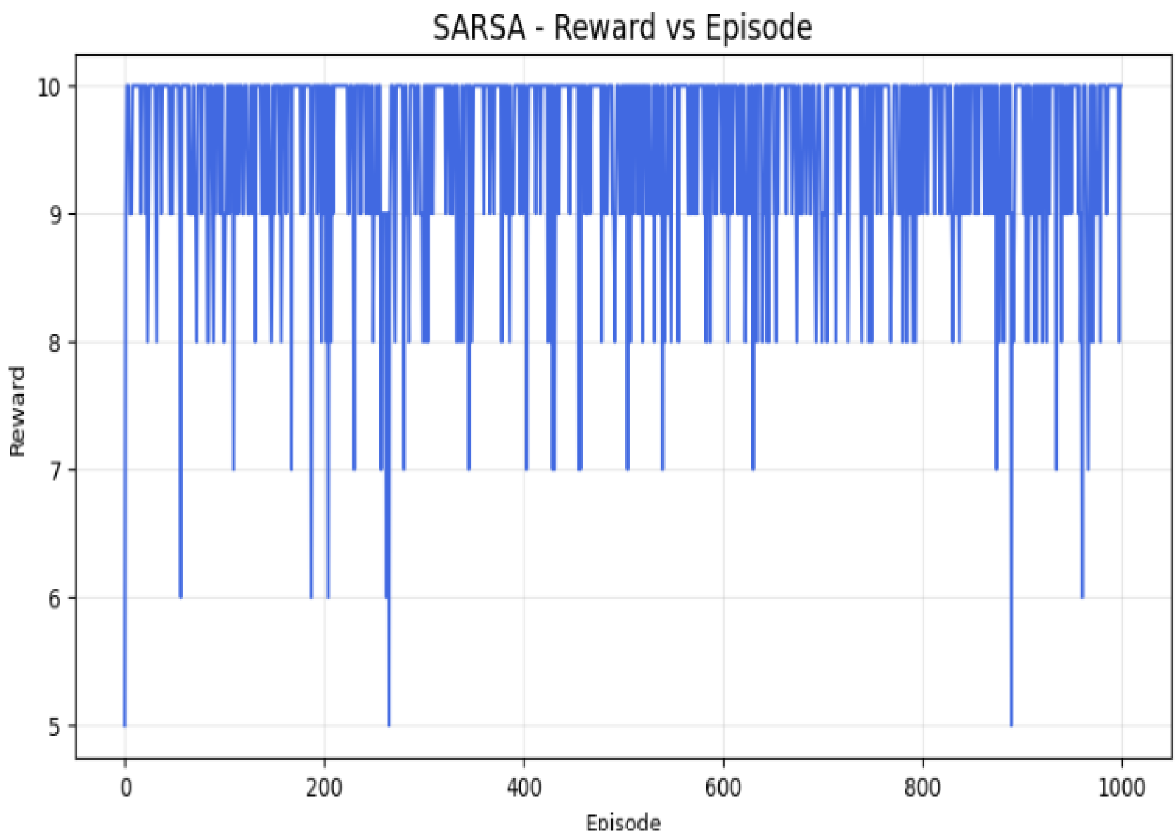
$$Q(St, At) \leftarrow Q(St, At) + \alpha[Rt + 1 + \gamma amaxQ(St + 1, a) - Q(St, At)]$$

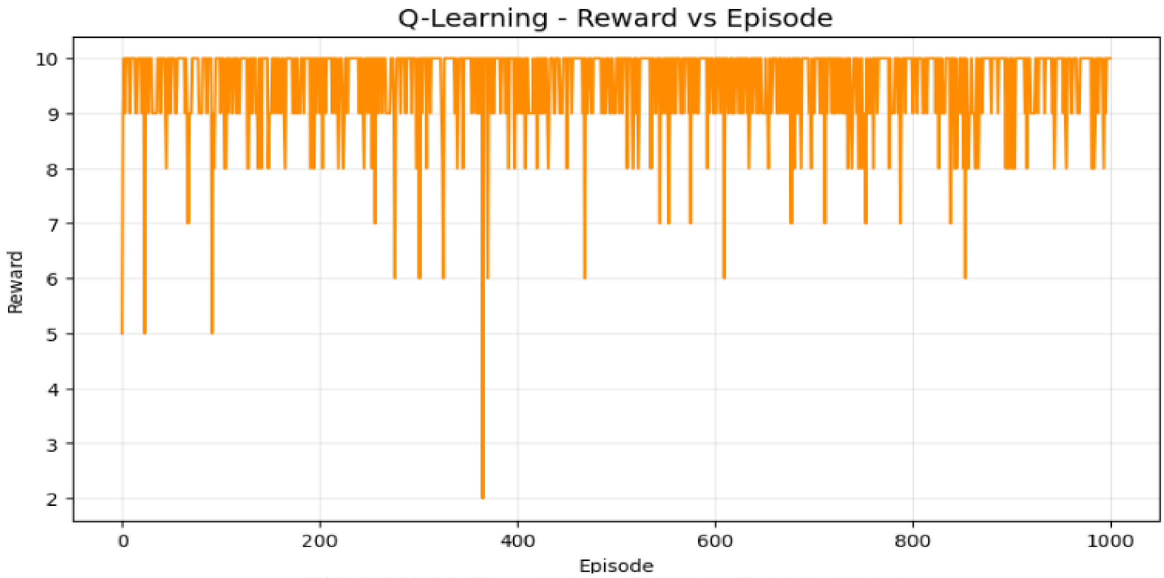
8. Repeat the training process for the specified number of episodes.
9. Record rewards, TD errors, steps, and success for both algorithms.
10. Extract the optimal policy using:

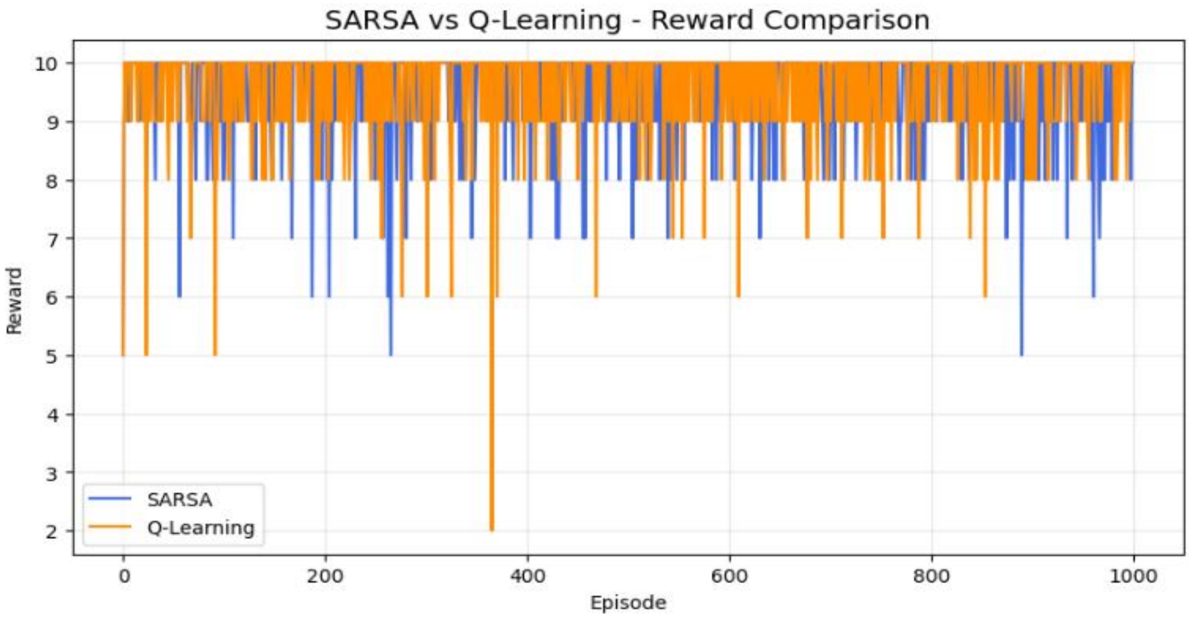
$$\pi * (s) = argamaxQ(s, a)$$

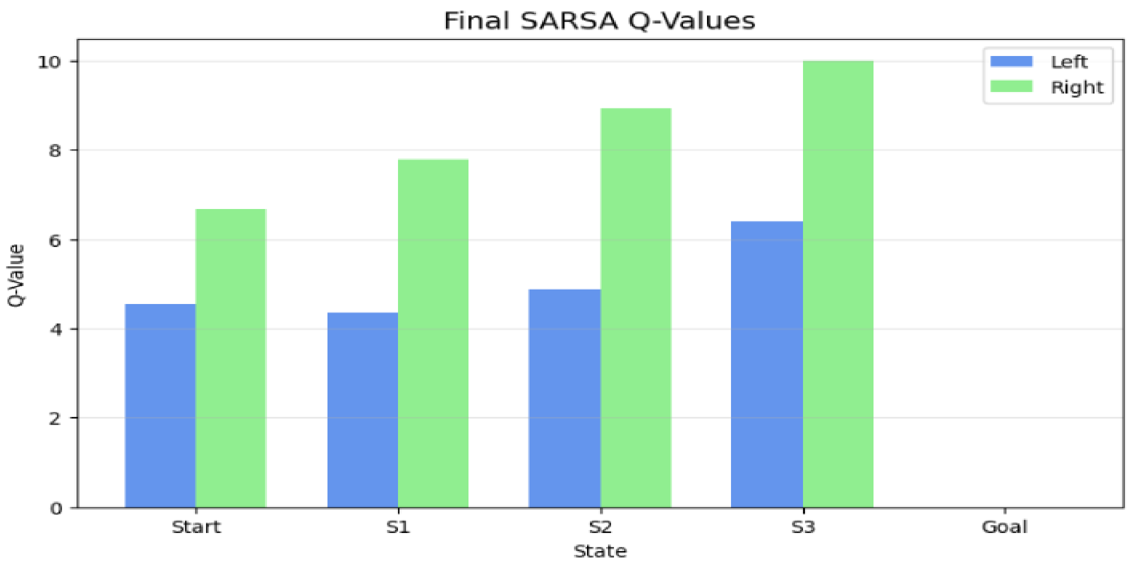
11. Compare the learned Q-tables and optimal policies.
12. Compare SARSA and Q-Learning using reward, success rate, TD error, episode length, and execution time.
13. Generate convergence and performance graphs.
14. Display summary tables and export the results to CSV files.
15. Analyze the difference between the **on-policy SARSA** and **off-policy Q-Learning** approaches.

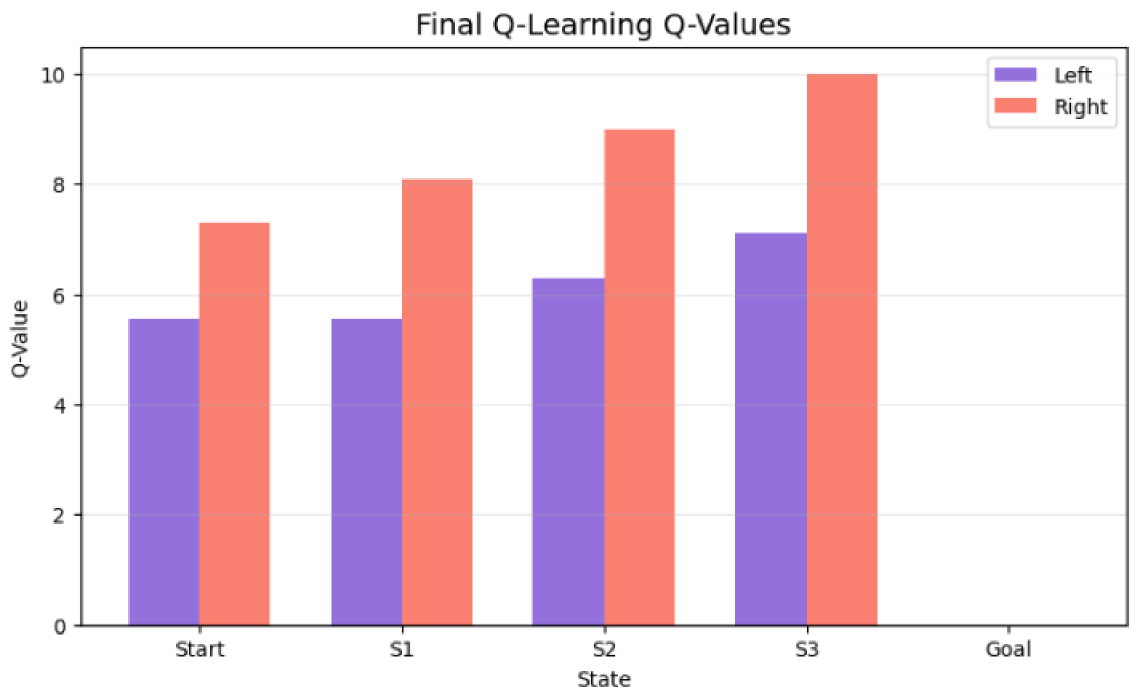
Visualization



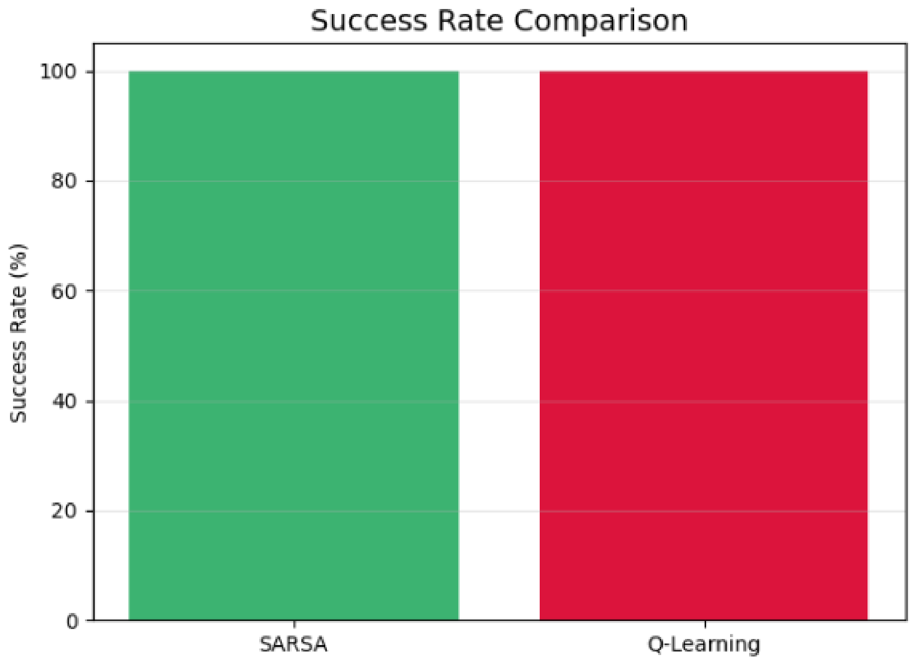

Q-Learning - Reward vs Episode
Reward
Episode
10
9
8
7
6
5
4
3
2
0
200
400
600
800
1000


SARSA vs Q-Learning - Reward Comparison
Reward
Episode
SARSA
Q-Learning
10
9
8
7
6
5
4
3
2
0
200
400
600
800
1000


Final SARSA Q-Values
Q-Value
State
Left
Right
Start
S1
S2
S3
Goal
10
8
6
4
2
0



Final Q-Learning Q-Values
Left
Right
Q-Value
10
8
6
4
2
0
Start
S1
S2
S3
Goal
State



Success Rate Comparison
Success Rate (%)
100
80
60
40
20
0
SARSA
Q-Learning

Summary Table – Q-Values and Policy

State	SARSA Left Q	SARSA Right Q	SARSA Optimal Action	Q-Learning Left Q	Q-Learning Right Q	Q-Learning Optimal Action
Start	Obtained	Obtained	Obtained	Obtained	Obtained	Obtained
S1	Obtained	Obtained	Obtained	Obtained	Obtained	Obtained
S2	Obtained	Obtained	Obtained	Obtained	Obtained	Obtained
S3	Obtained	Obtained	Obtained	Obtained	Obtained	Obtained
Goal	Obtained	Obtained	Terminal	Obtained	Obtained	Terminal

Result

SARSA and Q-Learning were successfully implemented and compared. SARSA learns using an **on-policy** approach, where the next action actually selected by the current policy is used for updating the Q-value. Q-Learning is an **off-policy** method that uses the maximum possible Q-value of the next state.